THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Rhizobium bacteria live in root nodules of: 1

(a) wheat **(b)** mango **(c)** rice **(d)** legumes (pulses)

2. In a starch test, parts of a variegated leaf that were white show: 1

(a) no blue-black colour **(b)** blue-black colour
(c) red colour **(d)** green colour

3. Photosynthesis occurs mainly in the: 1

(a) chloroplasts **(b)** nucleus **(c)** xylem **(d)** root cap

4. A plant which has no chlorophyll cannot: 1

(a) grow in size **(b)** respire
(c) take in water **(d)** make its own food

5. Pot C kept in the dark with water shows poor growth because of lack of: 1

(a) water **(b)** air **(c)** sunlight **(d)** soil

6. A kaleidoscope makes patterns by: 1

(a) refraction **(b)** absorption
(c) multiple reflections **(d)** scattering of sound

7. A shiny polished surface produces: 1

(a) absorbed light **(b)** regular reflection
(c) only shadows **(d)** no reflection

8. A torch is held close to a wall and a hand placed in between. The shadow: 1

(a) is transparent **(b)** is tiny
(c) does not form **(d)** is large and dark

9. Two plane mirrors parallel to each other form: 1

(a) many images **(b)** one image **(c)** no image **(d)** only shadow

10. A transparent object forms: 1

(a) luminous glow **(b)** a dark shadow **(c)** a clear image **(d)** almost no shadow

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What is the function of root hairs? 2

12. Why do plants kept in the dark die slowly? 2

13. What is the difference between regular and diffused reflection? 2

14. Why is a larger pinhole not preferred? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What is the role of the petiole and veins in a leaf? 3

16. Why can we not see around a corner directly? 3

17. Explain why plants are called autotrophs. 3

OR

Give reasons: 3
(a) leaves are green;
(b) root hairs are thin.

18. Explain with an example how the shape of a shadow can differ from the object. 3

OR

State the importance of light in our lives. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) How do roots absorb? 5
(ii) Role of root hairs.
(iii) Why mineral uptake matters.

OR

(i) Why is photosynthesis the base of food chains? 5
(ii) Its effect on air.
(iii) Role of forests.

20. (i) List the properties of light. 5
(ii) Explain two with activities.
(iii) Importance.

OR

(i) Why does light need no medium? 5
(ii) How does Sun's light reach us?
(iii) Compare with sound.

END OF QUESTION PAPER — Total: 40 Marks

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